

MACT 4233: Applied Multivariate Analysis

**Outlier Detection and Discriminant
Analysis of the Glass Identification Data**

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Dataset: Glass Identification Database

Methods: BACON, FLDA, FLDA2, and Multinomial Classification

Observations Used: 214 complete cases

Modeling Variables: RI, Na, Mg, Al, Si, K, Ca, and Ba

1 Introduction and Statement of the Problem

This report applies multivariate outlier detection and discriminant analysis to the Glass Identification dataset. The central question is how accurately forensic glass types can be classified from their measured chemical composition, and how much BACON-detected outliers affect the resulting classification rules. The report follows the same overall sequence as the attached template report: data description, pairwise visualization, BACON on the pooled data and within each class, normality assessment, FLDA, the special FLDA2 case with $p = 2$ and $k = 2$, multinomial classification, and a comparison before and after removing outliers.

The dataset is a classic forensic classification problem. Each sample is a piece of glass described by refractive index and elemental concentrations recorded as oxide weight percentages. In crime-scene work, fragments recovered from a scene can become evidence only if they are assigned to the correct source class. That makes the dataset a natural fit for discriminant analysis: the classes are meaningful, the predictors are continuous, and classification error has a real scientific interpretation.

One adjustment was necessary before reproducing the full analysis sequence. The smallest observed class, tableware, contains only 9 observations, while the raw feature set contains 9 quantitative variables. Standard within-class BACON requires $n > p$, so Fe was excluded from the multivariate modeling stage and retained only in the descriptive dataset documentation. This leaves 8 modeled variables and makes the BACON and class-specific covariance calculations well-defined for every observed class.

The scientific question of interest is whether the eight retained quantitative measurements contain enough information to separate the six observed glass types reliably, and whether removing chemically unusual observations materially improves classification performance.

2 Data Description

The data come from the Glass Identification Database. The working dataset contains 214 observations, no missing values, and 11 recorded variables: one identifier, nine continuous measurements, and one nominal class variable. Type 4 is not represented in the data, so the observed class structure consists of six categories: Types 1, 2, 3, 5, 6, and 7.

Table 1: Variables recorded in the Glass Identification dataset

Variable	Description	Type	Unit
Type	Glass type (class variable)	Nominal	Category
Id	Record identifier	Identifier	-
RI	Refractive index	Continuous	Unitless
Na	Sodium	Continuous	Weight % oxide
Mg	Magnesium	Continuous	Weight % oxide
Al	Aluminum	Continuous	Weight % oxide
Si	Silicon	Continuous	Weight % oxide
K	Potassium	Continuous	Weight % oxide
Ca	Calcium	Continuous	Weight % oxide
Ba	Barium	Continuous	Weight % oxide
Fe	Iron	Continuous	Weight % oxide

The class distribution is: Building windows (float): 70, Building windows (non-float): 76, Vehicle windows (float): 17, Containers: 13, Tableware: 9, Headlamps: 29. The discriminant analysis in the remainder of the report uses RI, Na, Mg, Al, Si, K, Ca, and Ba. Fe is omitted from modeling for the $n > p$ reason already stated and because it is the least variable measurement in the dataset. The analytic question is whether these eight retained variables are sufficient for accurate six-class classification and whether outlier removal improves the result.

3 Outlier Detection Analysis

3.1 Colored pairwise plot

Figure 1 shows a selected pairwise plot for Na, Mg, Al, and Ba, the four variables with the clearest class-separation signal in the data. Several patterns stand out immediately. First, the three window-glass classes (Types 1, 2, and 3) overlap substantially, especially in Na, Mg, and Al, so most classification difficulty is expected to occur inside that window-glass block. Second, the non-window classes are more distinctive: headlamps tend to occupy higher Ba and lower Mg regions, while containers tend to combine low Mg with relatively high Al. Third, the strongest relationships among the plotted variables are moderate rather than extreme: Mg is negatively related to Ba ($r = -0.49$) and Al ($r = -0.48$), while Ba and Al are positively related ($r = 0.48$).

Taken together, the figure suggests a mixed problem. Some class pairs look almost linearly separable, but the dataset as a whole is not dominated by one clean linear partition. That already hints that the six-class analysis will be much harder than the penguins example used in the template report.

Figure 10 displays a 4x4 grid of scatter plots showing the relationship between elemental ratios (Na, Mg, Al, Ba) and various object classes. The diagonal elements show the correlation coefficient (r) for each element: Na (0.16), Mg (-0.27), Al (0.48), and Ba (0.33). The off-diagonal plots show scatter plots of the ratios for each element, with data points colored by object class: T1 (blue), T2 (orange), T3 (green), T5 (red), T6 (purple), and T7 (brown). The x-axis for each column is labeled with the element name (Na, Mg, Al, Ba) and the y-axis for each row is labeled with the element name (Na, Mg, Al, Ba).

3.2 BACON on the full dataset

In the final iteration shown for reference in Figure 2, the BACON cutoff is 5.7507 and 70 observations lie above it. Importantly, these same 70 observations appear in every state of the cycle, so they form a persistent core of pooled outliers even though the algorithm as a whole does not converge. They are concentrated in

Type 2 and in the non-window classes, especially Types 5, 6, and 7. That pattern makes sense: BACON is trying to impose one pooled covariance structure on a dataset that actually contains several distinct chemical subpopulations.

The pooled BACON result is therefore interpreted descriptively rather than as the final outlier decision rule. The defensible outlier decisions for downstream classification come from the within-class BACON analysis in the next subsection, where each class is judged relative to its own chemical center and covariance structure.

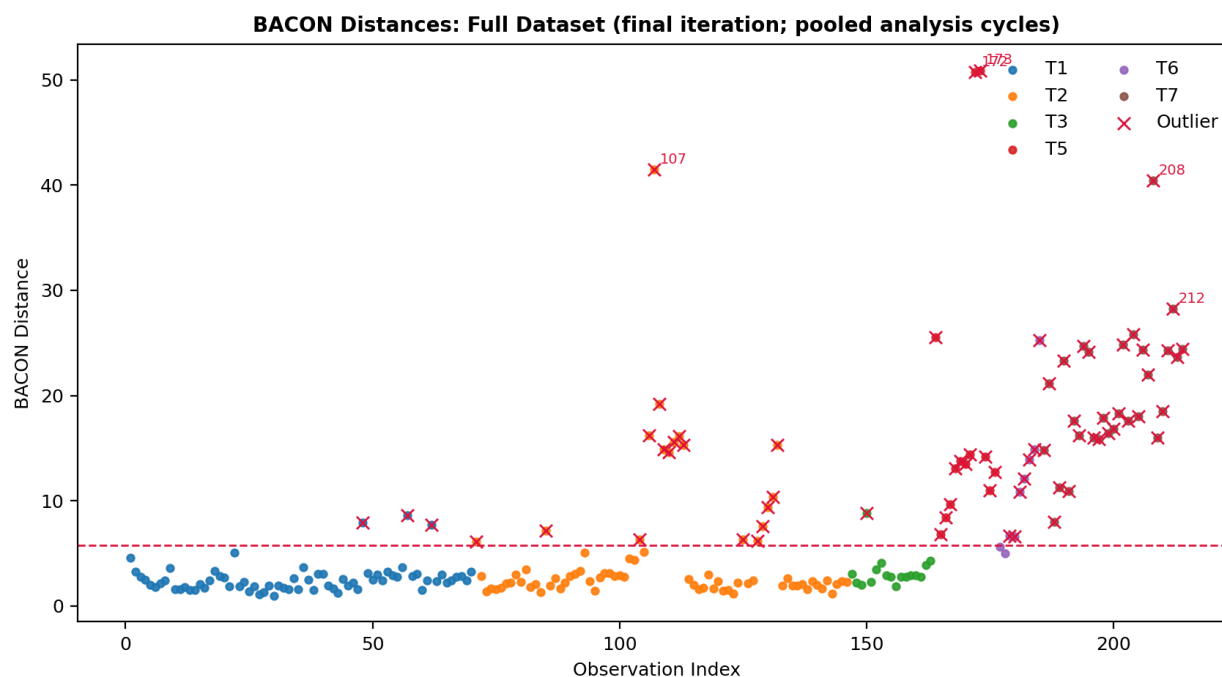


Figure 2: BACON distance index plot for the full dataset. The dashed horizontal line is the BACON cutoff; the pooled algorithm cycles instead of converging.

3.3 BACON within each class

Table 2: BACON results by observed glass type

Class	Group size	No. outliers	Cutoff	Outlier IDs
Type 1	70	10	6.1784	1, 22, 36, 48, 53, 54, 55, 56, 57, 62
Type 2	76	20	6.1080	71, 85, 93, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 125, 128, 129, 130, 131, 132
Type 3	17	0	9.6647	None
Type 5	13	3	13.5519	164, 165, 174
Type 6	9	0	46.5561	None
Type 7	29	8	9.5915	186, 187, 188, 189, 190, 202, 208, 212

Within-class BACON gives a much more interpretable result. No outliers were detected in Type 3 (vehicle windows, float) or Type 6 (tableware). In contrast, Type 1 contains 10 outliers, Type 2 contains 20, Type 5 contains 3, and Type 7 contains 8. The union of all within-class BACON outliers is 41 observations, so the cleaned dataset used later in the report contains 173 cases.

The class-specific patterns are informative. Type 2 outliers form the largest block and differ sharply from the remaining Type 2 observations, especially through much lower Mg and much higher Ca and Ba. Type 7 outliers shift toward higher Mg and K and away from the main headlamp profile. Type 5 shows only three outliers, but one of them is extremely far from the class center, which is why the Type 5 BACON plot has a very large vertical scale. Figure 3 highlights the within-class BACON outliers on the selected pairwise plot, and Figures 4 through 9 show the BACON distance plots for each class separately.

Selected Pairwise Plot with Within-Class BACON Outliers Highlighted

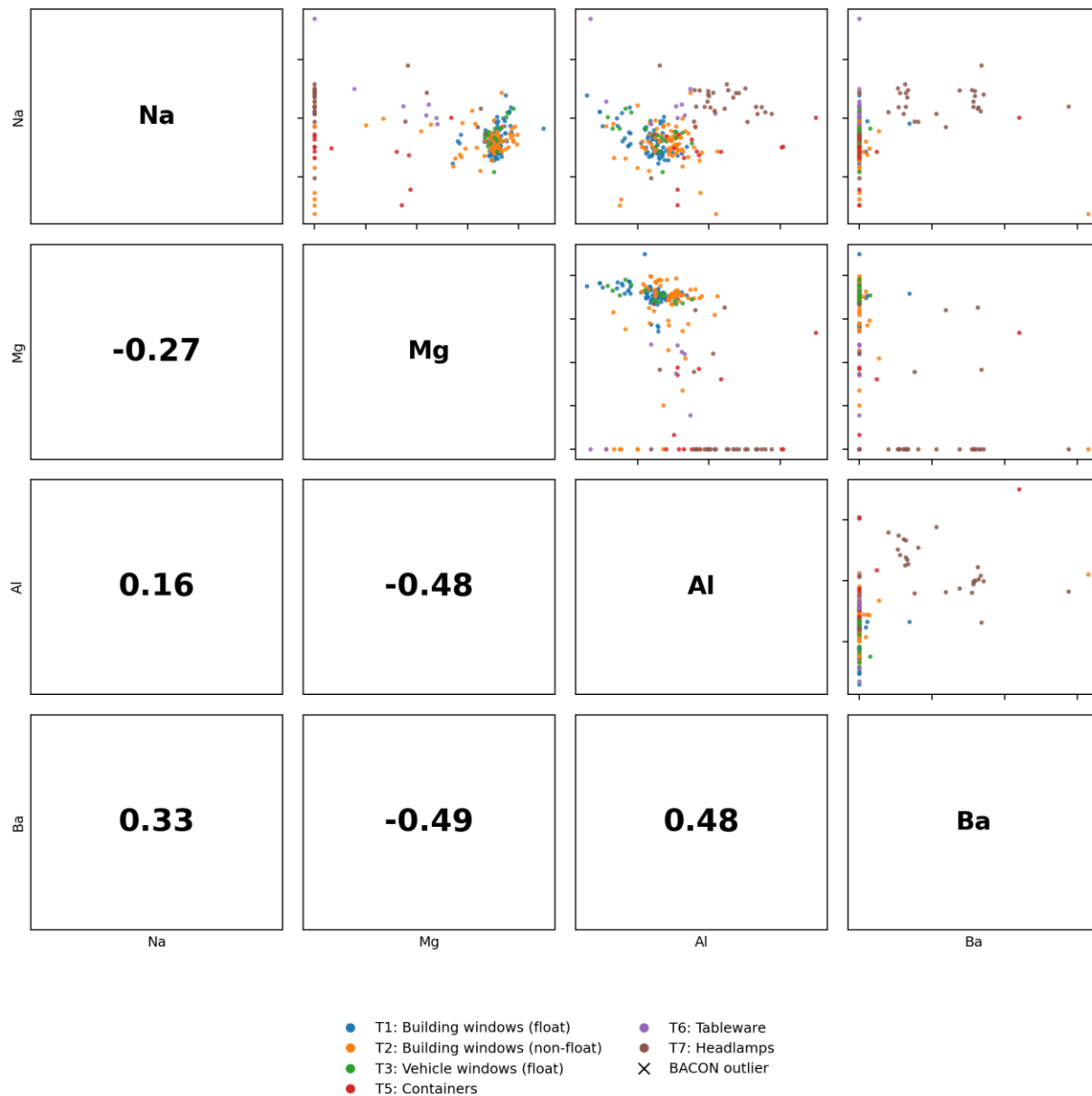


Figure 3: Selected pairwise plot with within-class BACON outliers highlighted.

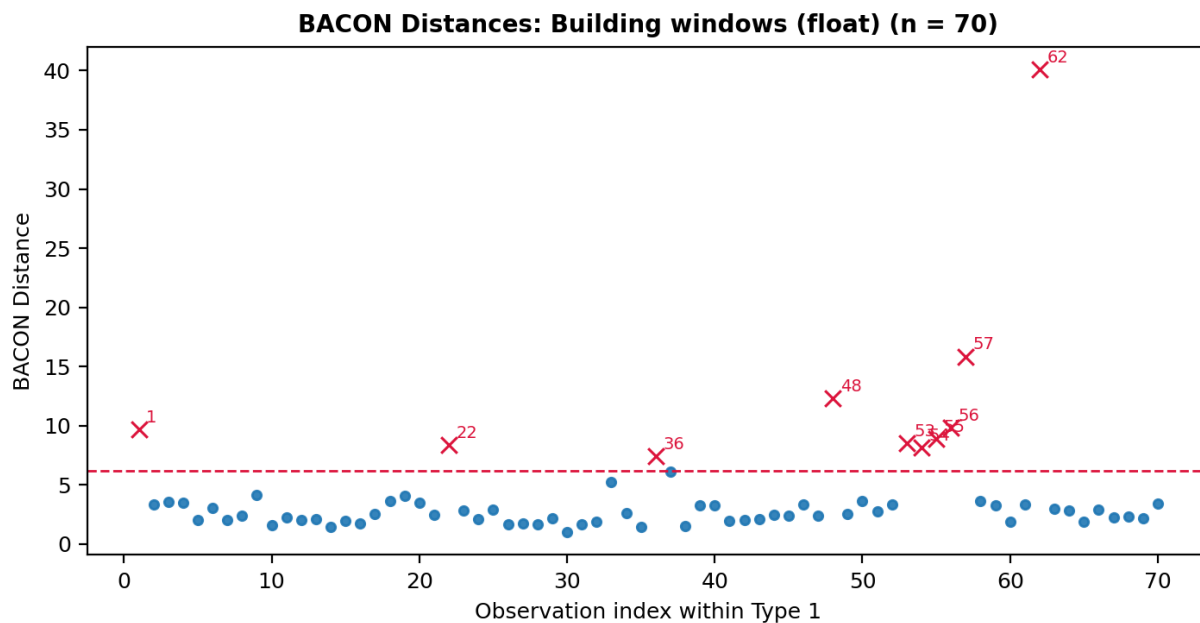


Figure 4: BACON distance plot for Building windows (float) ($n = 70$). Outlying IDs are labeled.

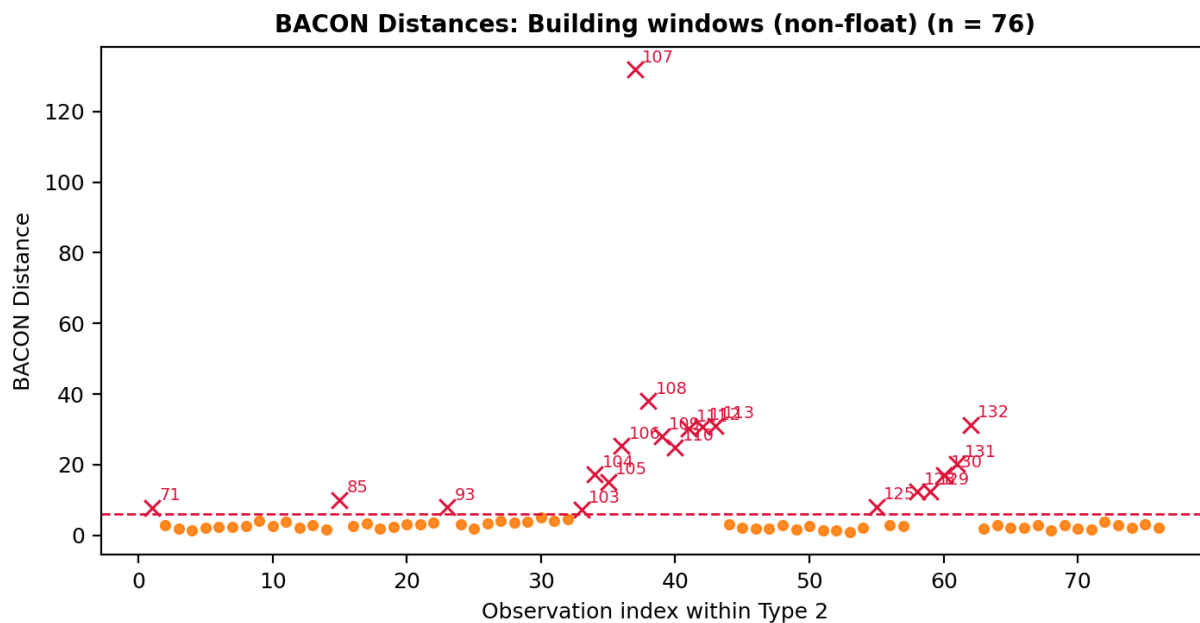


Figure 5: BACON distance plot for Building windows (non-float) ($n = 76$). Outlying IDs are labeled.

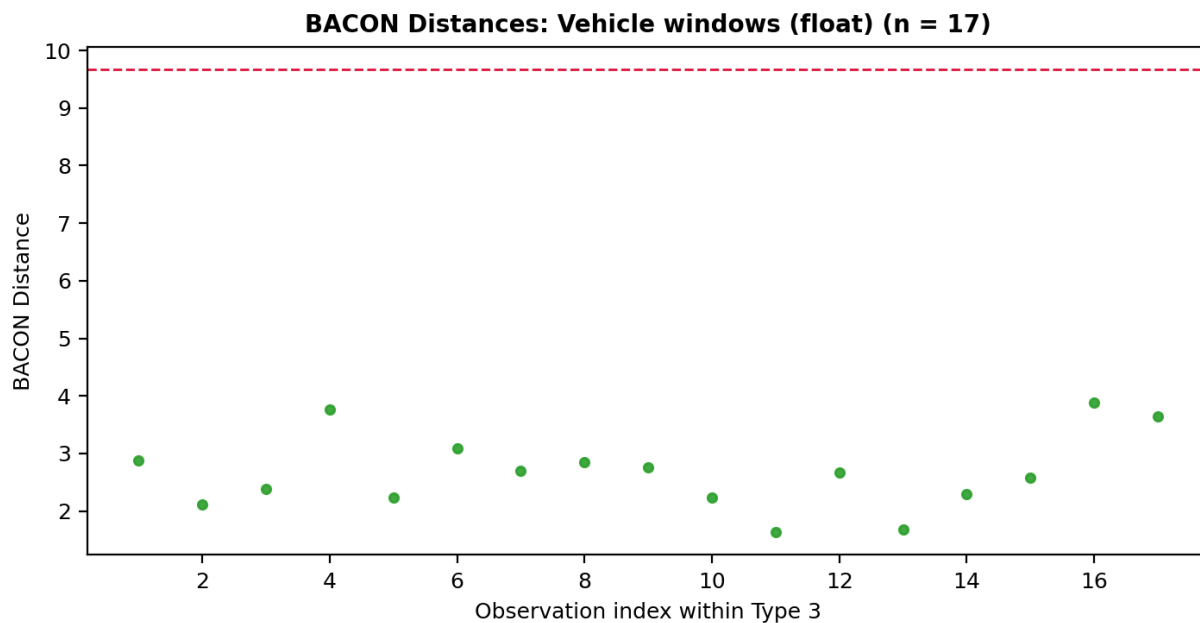


Figure 6: BACON distance plot for Vehicle windows (float) ($n = 17$). No observation exceeds the cutoff.

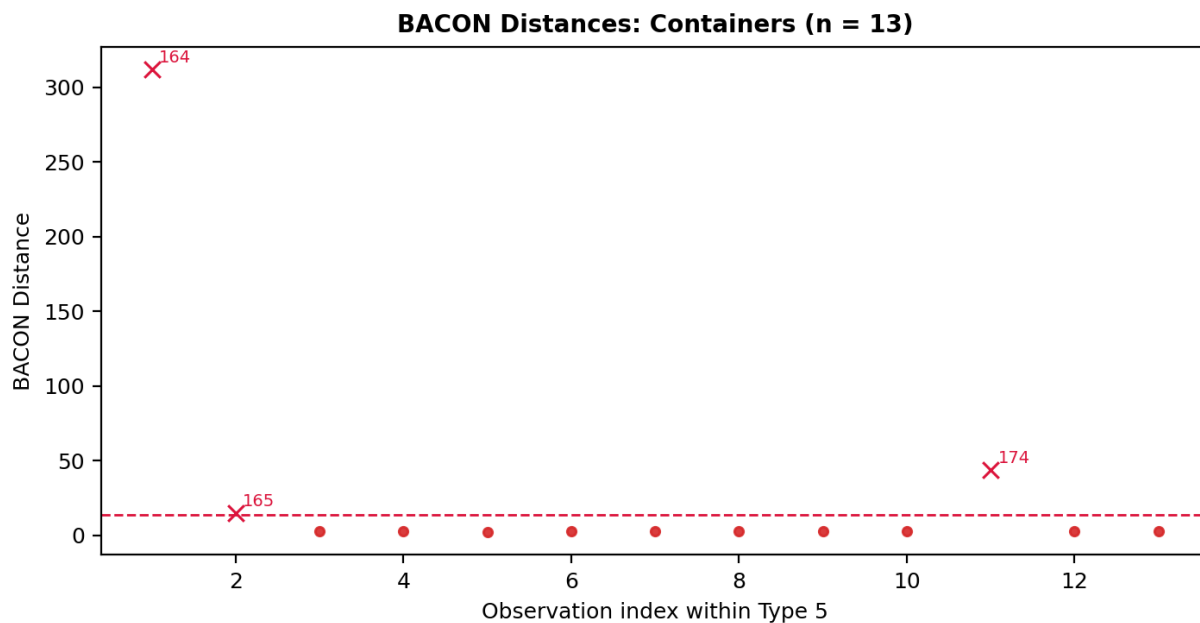


Figure 7: BACON distance plot for Containers ($n = 13$). Outlying IDs are labeled.

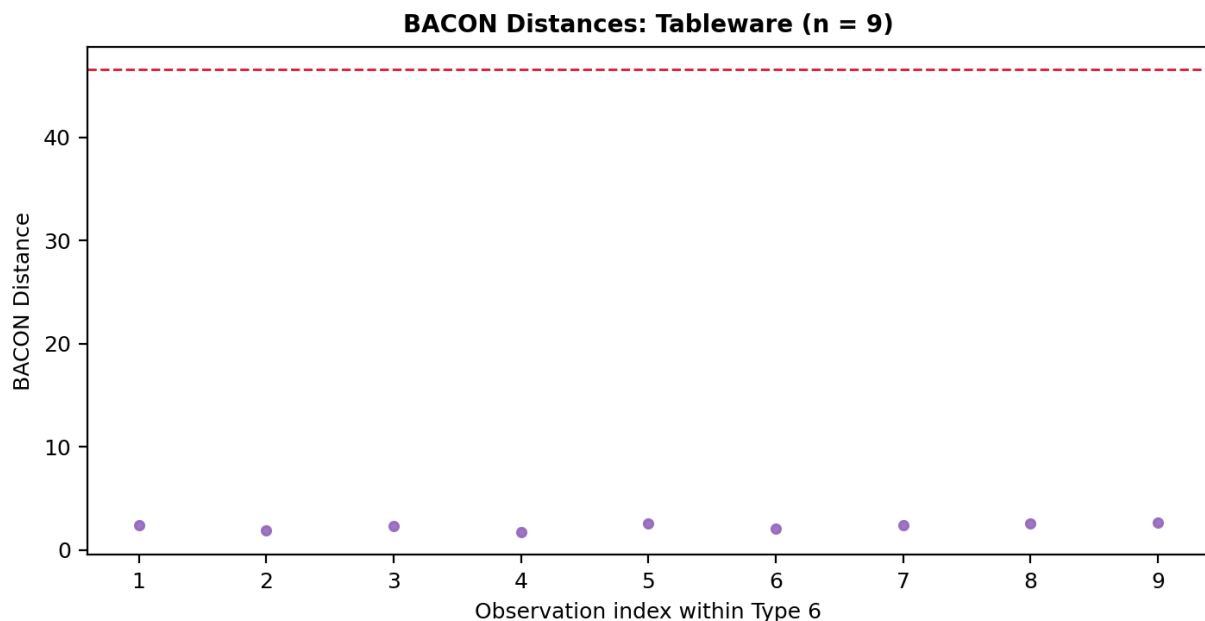


Figure 8: BACON distance plot for Tableware ($n = 9$). No observation exceeds the cutoff.

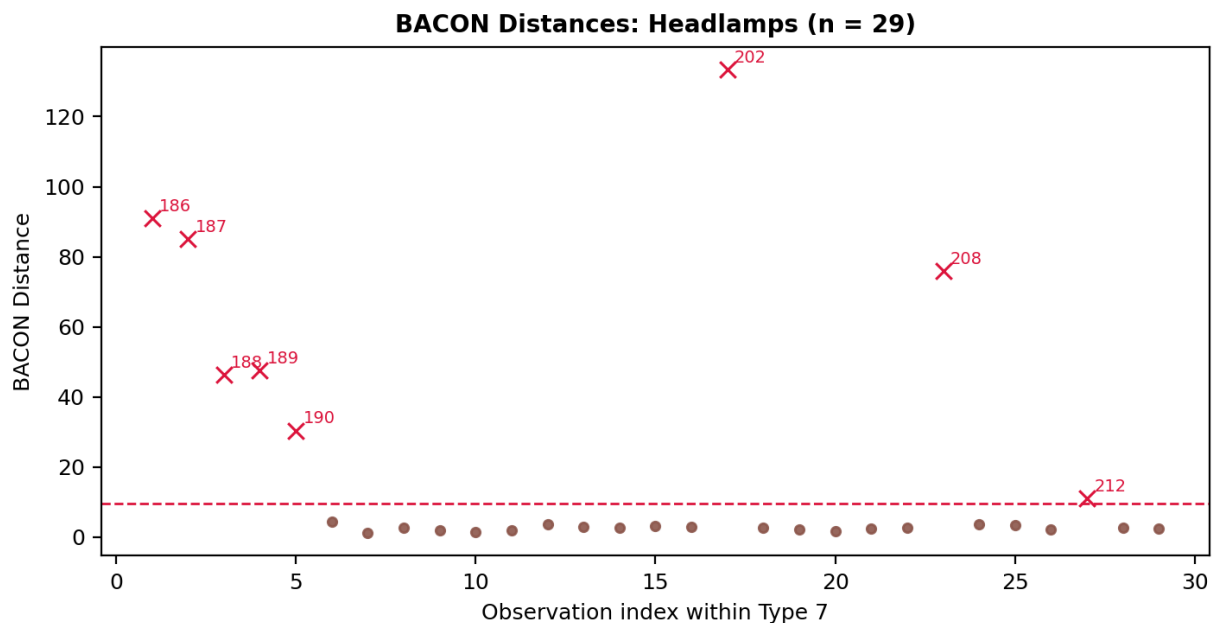


Figure 9: BACON distance plot for Headlamps ($n = 29$). Outlying IDs are labeled.

3.4 Conclusion of Outlier Detection

The outlier analysis leads to two conclusions. First, the pooled BACON analysis is not trustworthy as a final decision rule because the six-class mixture structure makes the one-cloud BACON assumption fail; the algorithm cycles and never settles. Second, once BACON is applied within each class, the results are stable and interpretable. The chemically clean classes are Types 3 and 6, while Types 1, 2, 5, and 7 contain substantial within-class heterogeneity. Those 41 within-class outliers define the cleaned dataset used in Section 5.

4 Discriminant Analysis

4.1 Normality check and scaling

Figure 10 shows Q-Q plots for the eight modeled quantitative variables. Normality is not close to perfect. Mg, K, and Ba show especially strong departures because many observations are exactly zero or clustered near zero, creating flat steps and heavy right tails. Ca also has a noticeable upper-tail departure. RI and Na are better behaved but still not perfectly normal. The multivariate normality assumption underlying FLDA is therefore only approximate and should be treated as a modeling convenience rather than a literal description of the data-generating process.

Even with that limitation, the template analysis can still be carried out so long as the limitation is acknowledged. As in the reference report, equal covariance matrices are assumed across classes for FLDA. Because the variables are measured on very different scales, all classification models were fit on standardized predictors.

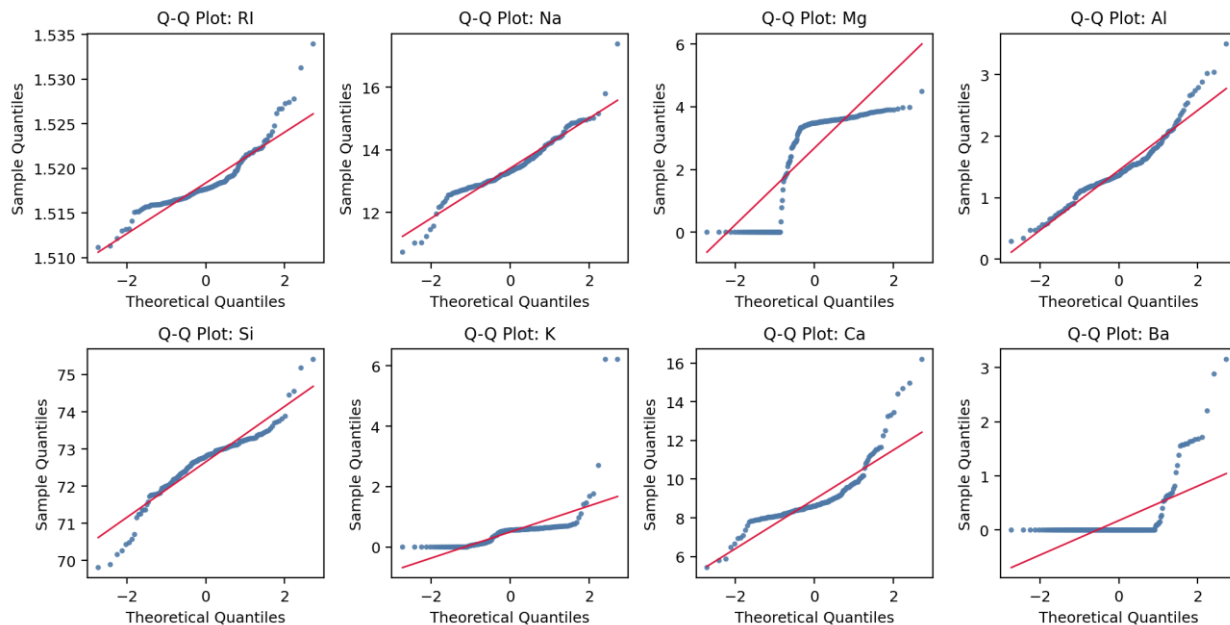


Figure 10: Q-Q plots for the eight quantitative variables used in the multivariate analysis.

4.2 FLDA: internal and external validation

FLDA performs much worse here than it did in the penguins report. The internal misclassification rate on the full data is 32.71%, and leave-one-out validation increases it to 35.05%. The reason is clear from the confusion matrices in the appendix: Types 1, 2, and 3 overlap heavily, and Type 3 is absorbed almost completely into Types 1 and 2. By contrast, Types 5, 6, and 7 are classified much more reliably.

So the story is not that FLDA fails everywhere; it fails exactly where the chemistry says it should fail: inside the overlapping window-glass family. That is the main structural weakness in this dataset.

4.3 FLDA2 for $p = 2$, $k = 2$

For the special FLDA2 case, the analysis was restricted to Type 1 (building windows, float) and Type 5 (containers), using scaled Mg and scaled Al. This two-class, two-variable view produces near-perfect

separation. The apparent error rate is 0.00%, and the leave-one-out error rate is only 1.20% (1 error out of 83 observations).

This is a useful stress test because it proves the problem is not that the dataset is inherently unclassifiable. Some class pairs are extremely easy. The real challenge is the overlap among the window-glass types when all six observed classes are analyzed simultaneously.

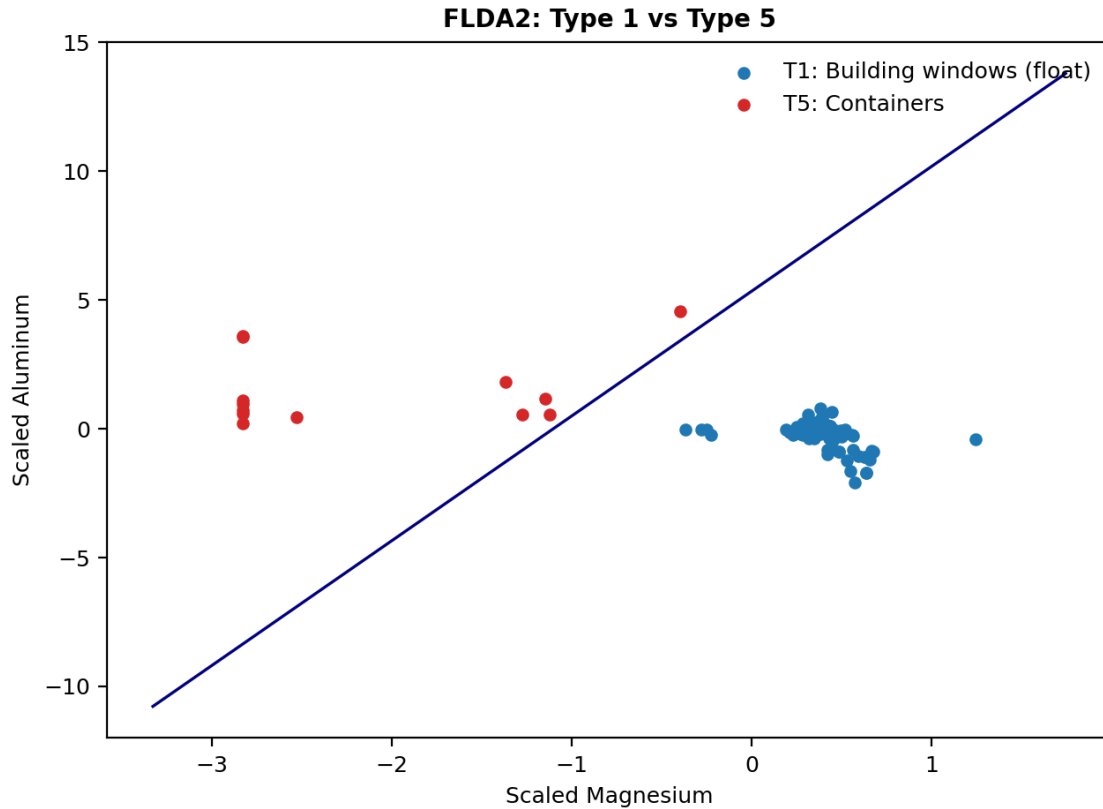


Figure 11: FLDA2 for Type 1 versus Type 5 using scaled magnesium and scaled aluminum.

4.4 Multinomial classification: internal and external validation

The multinomial model does not rescue the difficult six-class problem. On the full data, its internal misclassification rate is 33.18%, and the leave-one-out rate rises to 37.38%. That is slightly worse than FLDA on both validation schemes. The main confusion pattern is again concentrated in Types 1, 2, and 3; the model has no real ability to separate Type 3 from the other window-glass categories.

So there is no point pretending that the multinomial model is some magic fix. On this dataset it is not. The six-class window-glass overlap is the binding constraint, not the specific choice between linear discriminant analysis and multinomial logistic regression.

4.5 Comparison of methods before removing outliers

Table 3: Method comparison before and after removing within-class BACON outliers

Method	Internal (Full)	LOO (Full)	Internal (Clean)	LOO (Clean)
FLDA	32.71%	35.05%	19.65%	24.86%
Multinomial	33.18%	37.38%	21.39%	27.17%

Before removing outliers, FLDA has a small but consistent advantage over the multinomial model, both internally and under leave-one-out validation. Neither method is good enough to brag about on the full data, but FLDA is the less bad of the two. That is the honest reading of the numbers.

5 Effect of Removing the Outliers

After deleting the 41 within-class BACON outliers, the dataset size becomes 173. Repeating the classification analysis on this cleaned data produces a real improvement. FLDA drops from 32.71% to 19.65% internally and from 35.05% to 24.86% under leave-one-out validation. The multinomial model also improves materially, from 33.18% to 21.39% internally and from 37.38% to 27.17% under leave-one-out.

This is the opposite of the penguins case, where removing one outlier barely mattered. Here the outlier structure is not trivial, and removing within-class anomalies substantially tightens the classification problem. Even after cleaning, however, the six-class task remains hard because the remaining errors still come mainly from the overlap among Types 1, 2, and 3. Cleaning helps a lot, but it does not change the basic geometry of the window-glass classes.

6 Summary and Conclusions

This project analyzed the Glass Identification dataset using BACON outlier detection and discriminant analysis. The selected pairwise plot showed a mixed structure: several non-window classes are reasonably distinct, but the three window-glass classes overlap strongly. That overlap turns out to dominate the entire classification problem.

BACON behaved very differently in pooled and within-class form. On the pooled data the algorithm cycled, which is exactly what one should expect when forcing a single-cluster outlier detector onto a six-class chemical mixture. Within each class, however, BACON produced stable and interpretable results, flagging 41 outliers in total and leaving Types 3 and 6 unchanged.

The discriminant-analysis results are brutally clear. On the full data, neither FLDA nor the multinomial model classifies the six observed types particularly well, and the worst errors occur almost entirely among Types 1, 2, and 3. A two-class FLDA2 view shows that some class pairs are almost perfectly separable, so the difficulty is localized rather than universal.

Removing the within-class BACON outliers materially improves both methods. Even after cleaning, FLDA remains slightly stronger than the multinomial model, but neither method completely solves the window-glass overlap. The final conclusion is therefore: the dataset is only partially classifiable in its six-class form, pooled BACON is unstable because the data are genuinely multi-group, and within-class outlier removal helps a lot but does not eliminate the underlying class overlap.

A Appendix: Confusion Matrices

Rows represent the true class and columns represent the predicted class. The Class Error Rate column gives the proportion of observations in each true class that were misclassified.

Table 4: FLDA internal confusion matrix (full data, $n = 214$)

True class	T1	T2	T3	T5	T6	T7	Class Error Rate
T1	52	15	3	0	0	0	$18/70 = 25.71\%$
T2	17	54	0	3	2	0	$22/76 = 28.95\%$
T3	11	6	0	0	0	0	$17/17 = 100.00\%$
T5	0	5	0	7	0	1	$6/13 = 46.15\%$
T6	1	2	0	0	6	0	$3/9 = 33.33\%$
T7	1	2	0	1	0	25	$4/29 = 13.79\%$
Overall Error Rate							$70/214 = 32.71\%$

Table 5: FLDA leave-one-out confusion matrix (full data, $n = 214$)

True class	T1	T2	T3	T5	T6	T7	Class Error Rate
T1	51	16	3	0	0	0	$19/70 = 27.14\%$
T2	18	52	0	3	2	1	$24/76 = 31.58\%$
T3	11	6	0	0	0	0	$17/17 = 100.00\%$
T5	0	6	0	6	0	1	$7/13 = 53.85\%$
T6	1	2	0	0	5	1	$4/9 = 44.44\%$
T7	1	2	0	1	0	25	$4/29 = 13.79\%$
Overall Error Rate							$75/214 = 35.05\%$

Table 6: Multinomial internal confusion matrix (full data, $n = 214$)

True class	T1	T2	T3	T5	T6	T7	Class Error Rate
T1	50	19	1	0	0	0	$20/70 = 28.57\%$
T2	20	52	0	1	3	0	$24/76 = 31.58\%$
T3	8	9	0	0	0	0	$17/17 = 100.00\%$
T5	0	5	0	7	0	1	$6/13 = 46.15\%$

T6	0	2	0	0	7	0	$2/9 = 22.22\%$
T7	1	1	0	0	0	27	$2/29 = 6.90\%$
Overall Error Rate							$71/214 = 33.18\%$

Table 7: Multinomial leave-one-out confusion matrix (full data, $n = 214$)

True class	T1	T2	T3	T5	T6	T7	Class Error Rate
T1	49	20	1	0	0	0	$21/70 = 30.00\%$
T2	22	49	0	2	3	0	$27/76 = 35.53\%$
T3	8	9	0	0	0	0	$17/17 = 100.00\%$
T5	0	7	0	5	0	1	$8/13 = 61.54\%$
T6	0	2	0	0	6	1	$3/9 = 33.33\%$
T7	1	2	0	1	0	25	$4/29 = 13.79\%$
Overall Error Rate							$80/214 = 37.38\%$

Table 8: FLDA internal confusion matrix (clean data, $n = 173$)

True class	T1	T2	T3	T5	T6	T7	Class Error Rate
T1	46	12	2	0	0	0	$14/60 = 23.33\%$
T2	10	42	4	0	0	0	$14/56 = 25.00\%$
T3	2	3	12	0	0	0	$5/17 = 29.41\%$
T5	0	0	0	10	0	0	$0/10 = 0.00\%$
T6	1	0	0	0	8	0	$1/9 = 11.11\%$
T7	0	0	0	0	0	21	$0/21 = 0.00\%$
Overall Error Rate							$34/173 = 19.65\%$

Table 9: FLDA leave-one-out confusion matrix (clean data, $n = 173$)

True class	T1	T2	T3	T5	T6	T7	Class Error Rate
T1	44	12	4	0	0	0	$16/60 = 26.67\%$
T2	11	40	5	0	0	0	$16/56 = 28.57\%$
T3	2	5	10	0	0	0	$7/17 = 41.18\%$
T5	0	0	0	10	0	0	$0/10 = 0.00\%$
T6	1	0	0	2	6	0	$3/9 = 33.33\%$

T7	0	0	0	0	1	20	$1/21 = 4.76\%$
Overall Error Rate							$43/173 = 24.86\%$

Table 10: Multinomial internal confusion matrix (clean data, $n = 173$)

True class	T1	T2	T3	T5	T6	T7	Class Error Rate
T1	49	9	2	0	0	0	$11/60 = 18.33\%$
T2	12	43	1	0	0	0	$13/56 = 23.21\%$
T3	5	8	4	0	0	0	$13/17 = 76.47\%$
T5	0	0	0	10	0	0	$0/10 = 0.00\%$
T6	0	0	0	0	9	0	$0/9 = 0.00\%$
T7	0	0	0	0	0	21	$0/21 = 0.00\%$
Overall Error Rate							$37/173 = 21.39\%$

Table 11: Multinomial leave-one-out confusion matrix (clean data, $n = 173$)

True class	T1	T2	T3	T5	T6	T7	Class Error Rate
T1	46	11	3	0	0	0	$14/60 = 23.33\%$
T2	15	40	1	0	0	0	$16/56 = 28.57\%$
T3	5	10	2	0	0	0	$15/17 = 88.24\%$
T5	0	0	0	10	0	0	$0/10 = 0.00\%$
T6	1	0	0	0	7	1	$2/9 = 22.22\%$
T7	0	0	0	0	0	21	$0/21 = 0.00\%$
Overall Error Rate							$47/173 = 27.17\%$